

FutureProof EPM

Income from Property Equity — Technical Architecture

A comprehensive guide to the platform architecture, agents, workflows, and compliance systems that power monthly income distribution for equity-partnered homeowners.

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1. EPM Overview

FutureProof's Income from Property Equity solution is a fund-based income product that transforms home equity into sustainable monthly income. The model connects four stakeholders through a sophisticated distribution system powered by monthly rebalancing and performance monitoring.

The Four Stakeholders

Stakeholder	Role	Capital Flow
Customer	Homeowner who partners equity for monthly income	Receives monthly distributions from fund returns
Investment Partner	Manages invested capital	Generates returns; manages fund performance and rebalancing
Lender	Provides mortgage product & customer relationship	Retains margin on monthly distributions (0.75% to 2%)
Wholesale Funder	Supplies capital to lenders via funding lines	Receives wholesale margin on deployed capital (0.2% to 3.6%)

Monthly Distribution Flow

Fund Returns: Investment Partner's holdings generate returns aggregated and tracked monthly (on average 10% p.a. over 30 years).

Monthly Distribution: System calculates monthly payout amounts based on:

- Customer's equity percentage (typically 2% of home value)
- Fund performance month-to-date
- Interest holiday status (if triggered)

Amount to Customer: Customer receives their share of monthly returns as income.

Lender Margin: Lender deducts their margin from distribution (0.75% to 2% depending on tier).

Wholesale Margin: Wholesale Funder receives their cut (0.2% to 3.6% depending on structure).

FutureProof Margin: Platform takes its cut (0.25%) for ongoing management.

2. Customer Journey

PHASE 1: ORIGINATION (Weeks 1–4)

- Customer applies: Provides property details, equity percentage, income expectations through portal
- Deal Agent processes: KYC/AML checks, property valuation, risk tier assignment, document generation
- e-Signature: Customer reviews and signs loan documents, investment agreement
- Funding confirmation: Lender & Wholesale Funder approve. Capital transferred. Account activated.

PHASE 2: ONGOING SERVICING (Months 1–30+)

- Monthly distributions: Distribution Agent calculates returns every month. Customer receives payment on fixed date
- Performance monitoring: Investment Agent tracks fund performance. Generates rolling metrics visible on customer dashboard
- Interest holidays: If performance drops, holiday flag automatically triggered. Holiday status visible on customer statements and dashboard
- Statements & reports: Monthly statements show distribution amount, margins paid, fund performance, and holiday status
- Account management: Customer can request early exit, modify settings. Lender reviews requests within 5 business days

PHASE 3: EXIT (Early or Maturity)

- Customer-initiated: Request early repayment. Deal Agent calculates exit position (fund value minus outstanding fees)
- Lender approval: Lender reviews exit request, confirms repayment terms
- Settlement: Distribution Agent processes final payment. All accounts reconciled. Loan closed
- Documents archived: All transaction history, statements, audit logs preserved for 7 years. Compliance Agent completes final audit report

3. Stakeholder Account Views

Key information visible to each participant

Stakeholder	Key Account Views	Key Actions	Fraud Detection
Customer	• Monthly distribution • YTD cumulative income • Fund performance chart • Interest holiday status	• View transaction history • Download statements • Manage account settings • Request early exit	• Multiple withdrawals same day • Account access from unusual locations • Rapid device changes • Mismatch with known profile
Lender	• Monthly distribution schedule • Margin received • Customer payment status • Portfolio performance	• Approve interest holidays • View customer profiles • Adjust risk tier • Monitor portfolio health	• Systematic holiday requests • Unusual pattern changes • Distribution delays • Sudden customer exits
Investment Partner	• Fund performance metrics • ETF correlation data • Rebalancing schedule • Cost basis tracking	• Execute rebalancing • Manage ETF holdings • Report performance • Adjust hedges	• Unexplained underperformance • ETF holdings mismatches • Rebalancing delays • Cost basis discrepancies
Wholesale Funder	• Portfolio yield metrics • Default & recovery rates • Margin collections • Lender tier performance	• Approve lender tiers • Adjust funding lines • Initiate risk reviews • Approve new partners	• Yield compression • Concentration risk • Lender financial strain • Unusual default clustering

4. System Architecture

Core Platform

Rails 8.1 + Hotwire + Stimulus: Single monolithic Rails application with real-time updates via Hotwire (Turbo) for seamless user interactions and Stimulus controllers for client-side interactivity. Hotwire enables live dashboard updates without WebSocket complexity.

PostgreSQL Database: Single source of truth with full audit trails. Uses JSON columns for flexible metadata, materialized views for performance aggregation, database triggers for compliance rules. Partitioned by customer ID for scaling.

Kafka Event Queue: Asynchronous event streaming (distributions.*, holidays.*, compliance.*, deals.*). Ensures exactly-once processing with idempotent logic; dead-letter queues capture failures.

Redis Cache: Session management, rate limiting, computation caching. Bloom filters for fraud detection, cache-aside pattern for frequent access.

S3 Storage: Document archival with versioning and encryption. Lifecycle policies: archive after 90 days to Glacier, delete after 7 years.

Core Services

Distribution Engine: Runs 1st-5th monthly. Pulls fund returns, calculates distributions. Stages in Redis, executes ACH transfers via Stripe, publishes events. Retry logic with exponential backoff.

Compliance Engine: Real-time validation against 20+ regulatory rules. Audit trail via immutable event log. Change Data Capture via database triggers.

Fraud Detection: ML pipeline trained on historical patterns. Real-time scorer checks: (1) known patterns (Bloom filter), (2) behavioral anomalies (isolation forest), (3) rule-based checks.

Interest Holiday Logic: Monthly job calculates rolling returns, compares thresholds. State machine manages transitions.

Notification Service: Email (SendGrid), SMS (Twilio), in-app. Delivery guarantees with idempotent keys and confirmation tracking.

Document Engine: ERB templates + Prawn PDF. Generates contracts, agreements, statements. Digital signatures, S3 archival.

5. Four-Agent Architecture

Agent	Primary Role	Key Workflows	Triggers
Deal Agent	Customer acquisition & onboarding	• KYC/AML checks, property valuation • Risk tier assignment • Document generation & e-sign • Loan approval	• New application • Compliance clearance • Fund confirmation
Distribution Agent	Monthly returns calculation & payment	• Capital distributions • Deduct lender/funder margins • Execute payments • Generate statements • Handle retries	• Monthly cutoff • Fund performance updates • Holiday status changes
Investment Agent	Fund monitoring & interest holidays	• Monitor fund performance • Calculate rolling metrics • Determine holiday eligibility • Trigger/release holidays	• Monthly updates • Threshold breach • Recovery confirmation
Compliance Agent	Monitoring, enforcement & reporting	• Real-time fraud detection • Regulatory rule enforcement • Generate audit reports • Alert suspicious activity	• Transaction submitted • Fraud score threshold • Rule violation detected

Agent Communication: All agents publish events to Kafka; others subscribe and react asynchronously. Decouples workflows, enables real-time coordination. Deal Agent → deal.created → Distribution & Compliance subscribe. Distribution Agent → distribution.monthly → Lender/Funder receive. Investment Agent → holiday.triggered/released → Distribution adjusts. Compliance Agent → fraud.detected → Escalation to lender/admin.

Agent Technology Implementation

Framework: Rails service classes (DealAgent::OnboardingService) → Sidekiq background jobs (4 workers/pod) → Horizontally scalable. Kafka: Consumer groups, auto-partitioning, dead-letter queues. Idempotency: Database constraints + transactional locks. Retries: Exponential backoff (1s→2s→4s→8s→16s). Tracing: trace_id UUID tags all logs. State Machines: AASM gem (deal: draft→submitted→kyc_pending→approved→funded→active; holiday: normal→flagged→active→recovery→normal). Twilio Flex: Escalation dashboard, auto-call routing, recording, screen-pop. Sync/Async: Defaults async; sync for Stripe (5s), IDology (10s), valuation (15s) with circuit breakers.

6. Interest Holiday Mechanism

Interest holidays FLAG market conditions—they do NOT suspend payments. During a holiday, interest accumulates in the customer's account rather than being distributed.

Parameter	Value	Meaning
Entry Threshold	0.90 (90%)	If fund value drops to 90% of starting value, entry conditions met
Exit Threshold	1.458 (145.8%)	When fund recovers to 145.8%, holiday automatically released
Maximum Duration	Variable	Holiday expires if conditions not met within timeframe
Accumulated Interest	Ongoing	During holiday, interest not distributed but accumulates in account

Timeline Example: Months 1–10 normal operations (fund performing, customer receives distributions). Month 11: market downturn, fund drops to 88%, entry threshold (90%) breached, holiday flagged. Months 12–18: holiday active (interest accumulates, customer continues distributions, lender alerted). Month 19: recovery confirmed, fund reaches 145.8%, holiday removed, status returns normal.

7. Compliance & Fraud Detection

Compliance Rules

Rule	Validation	Failure Action
KYC/AML	Identity verified before onboarding	Application rejected; customer notified
Responsible Lending	Fund appropriate for customer profile	Risk tier adjusted; margin increased
Distribution Accuracy	Calculations verified; margins correct	Distribution held pending audit
Fund Performance Audit	Returns match investment partner reports	Discrepancy flagged; reconciliation initiated
Holiday Logic	Thresholds correctly applied	Holiday manually reviewed; investor notified
Data Privacy	PII access logged; encryption verified	Unauthorized access alert; investigation

Real-Time Fraud Monitoring

Account Access: Logins from unusual locations, rapid device changes, after-hours patterns. Transactions: Multiple withdrawals same day, requests immediately after distribution, systematic holiday manipulation. Data: Mismatches between reported/actual fund values, distribution errors, margin discrepancies. Violations: KYC/AML failures, undisclosed conflicts, unauthorized transfers. Alerts escalate to lenders/funders; critical incidents require manual review.

Audit & Reporting

Audit Trail: Every transaction, margin payment, holiday trigger recorded with timestamp, user ID, changes. Retained 7 years minimum. Monthly Reporting: Lenders/funders receive automated reports on portfolio health, margin collections, defaults. Annual Compliance: External auditor covers regulatory requirements across Australia, UK, US jurisdictions.

8. Database Schema

Core Tables

Table	Purpose	Key Fields
customers	Homeowners in partnership	id, email, kyc_status, aml_status, risk_tier
properties	Residential with equity partnerships	id, customer_id, address, property_value, equity_pct, loan_amount
accounts	Customer investment accounts	id, customer_id, starting_balance, current_balance, holiday_flagged
fund_performance	Monthly metrics (materialized view)	account_id, month, starting_value, ending_value, return_pct
distributions	Monthly payouts (immutable)	id, account_id, fund_return, margins, customer_payout, executed_at
payments	ACH transfers (immutable)	id, distribution_id, amount, bank_account_id, status, stripe_id
audit_log	Event log (append-only)	id, entity_type, action, changes_json, user_id, timestamp
fraud_alerts	Suspicious patterns	id, customer_id, alert_type, fraud_score, status
lenders	Loan originators	id, name, tier, max_funding_line, margin_rate
wholesale_funders	Capital providers	id, name, funding_capacity, deployed, margin_rate

Design Patterns

Immutable Logs: distributions, payments, audit_log INSERT-only with timestamps. Soft Deletes: customers, accounts include deleted_at. JSON Columns: changes_json, details_json for flexible schemas. Materialized Views: fund_performance refreshed nightly. Partitioning: distributions/payments by created_at (monthly). Indexes: (customer_id, created_at), (account_id, month), (status, created_at).

9. Technology Stack

Backend & Processing

- Rails 8.1 (Ruby 3.4.8)
- Sidekiq 7.x (4 workers)
- ruby-kafka (events)
- Puma (3 threads, auto-scaling)

Data & Persistence

- PostgreSQL 16 (primary + replica)
- Redis 7 (cache, sessions)
- Kafka 3.6 (3-replica topics)
- PgBouncer (pooling)

Agent & Workflow

- Sidekiq (async execution)
- Kafka (pub/sub)
- OpenTelemetry (tracing)
- Dead-letter queues
- State machines

Customer Communication

- SendGrid (email)
- Twilio (SMS + voice)
- Twilio Flex (call center)
- WebRTC (video)
- DocuSign (e-sig)

Frontend & UI

- Hotwire (Turbo + Drive)
- Stimulus JS
- Chart.js (viz)
- ERB templates

Storage & Infra

- AWS S3 (versioned)
- Fly.io (deployment)
- GitHub Actions (CI/CD)
- Docker
- Terraform

Monitoring & Security

- Sentry (errors)
- CloudWatch (logs)
- Grafana (dashboards)
- PagerDuty (alerts)
- HTTPS/TLS, CSP

10. Deployment & Quality Assurance

Testing Strategy

Unit Tests: Distribution calculations, holiday logic, fraud rules. Integration Tests: Agent workflows, customer journeys, payment processing. Performance Tests: Batch processing at scale. Security Tests: Auth, authorization, data access, encryption. Manual: Portal UX, reporting accuracy, admin workflows.

Continuous Deployment

GitHub Actions: RSpec (200+ specs), Rubocop, Brakeman, migrations validated on every PR. Staging: Auto-deployed from main, production-replica DB, manual smoke tests. Production: Blue-green on Fly.io, 10%→50%→100% canary over 10 minutes. Health checks: Rails boot, DB connection, Kafka reachable, <200ms latency. Auto-rollback if checks fail in 5 minutes.

Monitoring & Alerting

Sentry: Real-time error tracking, source maps, PagerDuty escalation <2 minutes. CloudWatch: Structured JSON logs (lograge), trace_id tagging, 90-day retention. Grafana: System health (CPU/memory/disk), app metrics (requests/sec, p95 latency), business metrics (distributions, payments, revenue), agent health (lag, dead-letter), DB health (connections, replication). Alerts: error rate >1%, distribution latency >60s, Kafka lag >1000 messages.

